GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-052 Routine Substation Inspection

Field	Value
Document ID	PR-ELEC-052
Version	4
Effective from	2024-03-04
Review due	2026-06-30
Owner	Head of Electricity Network Operations
Approved by	Senior Authorised Person (HV)
Applies to	All GigaWhat Electricity sites, plant and equipment

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets how often GigaWhat Electricity substations are inspected and what is checked, so that defects are found and put right before they cause danger or loss of supply.

2 Scope

This procedure applies to ESS-RVL, ESS-ASH, ESS-KGM, ESS-BRK and ESS-ORC. Routine inspection is visual only: no covers are removed and no HV switching is carried out. Entry into the Kingsmead cable tunnel CT-KGM-01 is covered by PR-ELEC-031.

3 References

- PR-ELEC-031 Entry to Kingsmead Cable Tunnel (Confined Space)
- PR-ELEC-044 Transformer Oil Sampling and Dissolved Gas Analysis
- PR-ELEC-047 Response to Transformer Temperature Alarms
- PR-ELEC-058 Substation Battery Maintenance
- PR-ELEC-063 Substation Access, Keys and Visitors
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- Electricity Safety, Quality and Continuity Regulations 2002

4 Definitions

Category A defect: a defect that causes immediate danger or risk of imminent failure.

Category B defect: a defect to be repaired within 28 days.

Category C defect: a defect to be repaired at the next planned maintenance.

Silica gel breather: the desiccant unit that dries air entering a transformer conservator.

5 Responsibilities

Senior Authorised Person (HV): sets the inspection programme and reviews every Category A and B defect.

Authorised Person (HV): reviews completed checklists each month and assigns defects for repair.

Competent Person (HV): carries out inspections and records findings.

Control Engineer: logs arrivals and departures and receives Category A defect reports.

6 Procedure

6.1 Inspection intervals

Site type	Sites	Interval
Grid substation	ESS-RVL	Monthly
Primary substation	ESS-ASH, ESS-KGM	Monthly
Secondary substation	ESS-BRK, ESS-ORC	Quarterly
Cable tunnel shaft heads (surface only)	CT-KGM-01	Monthly, with ESS-KGM

6.1.1 Carry out each inspection within 7 days either side of its due date.

6.1.2 Carry out an extra inspection after severe weather, flooding, a reported intrusion or a transformer temperature alarm.

6.2 Arrival, security and signs of intrusion

> WARNING: Intruders may have stolen earth conductors. Metalwork that has lost its earth connection can become live and give a fatal shock.

6.2.1 Inform the Control Engineer and sign the site log under PR-ELEC-063. Lone inspection is allowed with check-ins under PR-CORP-004.

6.2.2 Walk the perimeter before entering: check that fences, walls, gates and locks are intact and that danger signs are present and legible.

6.2.3 Look for signs of intrusion such as forced doors, broken locks, cut fencing, missing or cut earth conductors and disturbed ground.

6.2.4 If you find signs of intrusion or earth theft, do not enter. Withdraw and inform the Control Engineer, and call 999 if intruders may be present.

6.2.5 Check for vegetation or objects near the fence that could help someone climb it.

6.3 Transformers

6.3.1 Check the conservator oil level gauge is within the normal band for the current temperature.

6.3.2 Look for oil leaks from the tank, radiators, valves and gaskets, and check the bund is free of oil and water.

6.3.3 Check the silica gel breather. Replace the gel when more than two-thirds has changed from orange to green, and check the oil seal cup is filled to its mark.

6.3.4 Record the current OTI and WTI readings and the maximum pointer readings, then reset the maximum pointers.

6.3.5 Report any maximum pointer reading above the Stage 1 settings in PR-ELEC-047 to the Control Engineer.

6.3.6 Check that cooling fans are in AUTO, the Buchholz relay window is clear of gas and the pressure relief flag is not raised.

6.3.7 Check bushings and cable boxes for cracks, tracking or leaking compound, and listen for abnormal noise.

6.4 Switchgear and ring main units

6.4.1 Check that position indicators agree with the network diagram held by the Control Engineer.

6.4.2 Record circuit breaker operation counter readings.

6.4.3 On SWG-KGM-11, check the oil level gauges and look for oil seepage. Do not stay in front of the panels longer than needed.

6.4.4 On gas-insulated ring main units, check the gas pressure gauge is in the green band.

6.4.5 Listen and smell for discharge, such as crackling or an ozone smell, and look for tracking or condensation.

6.4.6 Check that switchroom heaters work and there is no water ingress.

6.5 Batteries, earthing and buildings

6.5.1 At ESS-RVL, ESS-ASH and ESS-KGM, record the battery float voltage, check the charger shows no alarms and check the battery room ventilation. Detailed checks are in PR-ELEC-058.

6.5.2 Check visible earth conductors and connections are intact, tight and free of corrosion.

6.5.3 Check roofs, doors, windows and cable trench covers for damage.

6.5.4 Check that fire extinguishers are in date and that first aid kits, emergency lighting and the site emergency information are in place.

6.6 Recording and reporting defects

6.6.1 Record all readings and findings on the inspection checklist for the site.

6.6.2 Report Category A defects to the Control Engineer at once, make safe if you can, and stay until the site is made safe or handed over.

6.6.3 Raise Category B and C defects in the works management system within 2 working days.

6.6.4 Report any incident or near miss under PR-CORP-001.

6.6.5 Sign out with the Control Engineer and leave the site secure under PR-ELEC-063.

7 Records

- Completed inspection checklists, kept for six years.
- Defect records in the works management system.
- Maximum pointer readings, copied to the transformer condition record.

8 Revision history

Version	Date	Change
1	2012-02-06	First issue.
2	2016-03-07	Added secondary substations at a quarterly interval.
3	2020-03-02	Added battery charger and ventilation checks.
4	2024-03-04	Added the silica gel replacement criterion, checks for intrusion and earth conductor theft, recording of maximum pointers and defect categories.